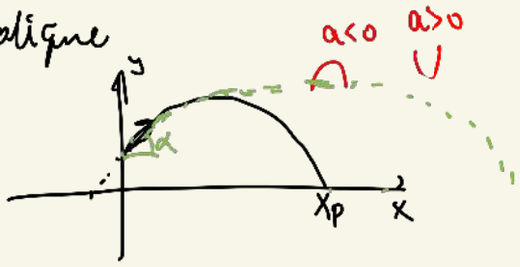


Ex6 : $a<0$ b c

8) $y = \frac{-g}{2V_0^2 \cos^2\alpha} x^2 + (\tan\alpha)\, x + h$ ←

9) parabolique



10) x_p.

$y(x_p) = 0$

$\Delta = b^2 - 4ac$

$= (\tan\alpha)^2 - 4 \cdot \left(\frac{-g}{2V_0^2 \cos^2\alpha}\right) h$

$= (\tan\alpha)^2 + \frac{2gh}{V_0^2 \cos^2\alpha}$

$x_p = \frac{-b - \sqrt{\Delta}}{2a} = \frac{-\tan\alpha - \sqrt{(\tan\alpha)^2 + \frac{2gh}{V_0^2\cos^2\alpha}}}{2 \cdot \left(\frac{-g}{2V_0^2\cos^2\alpha}\right)}$

$x_p = \frac{V_0^2 \cos^2\alpha \tan\alpha}{g} \cdot \left(1 + \sqrt{1 + \frac{2gh}{V_0^2 \cos^2\alpha \tan^2\alpha}}\right)$

$= \underbrace{\frac{V_0^2 \cos\alpha \sin\alpha}{g}}_{(*)} \cdot \left(1 + \sqrt{1 + \underbrace{\frac{2gh}{V_0^2 \sin^2\alpha}}_{(**)}}\right)$

11) $V_0 = 14\,m/s$, $\alpha = 42°$, $h = 2{,}2\,m$, $g = 9{,}81\,m/s^2$

$(*) = \frac{14^2 \cos 42° \sin 42°}{9{,}81} = 9{,}935\,m$

$(**) \frac{2 \cdot 9{,}81 \cdot 2{,}2}{14^2 \sin^2 42°} = 0{,}4919$

$x_p = 9{,}935 \cdot (1 + \sqrt{1{,}4919}) = 22{,}07\,m$

12) V_0 sera plus grand → a ↓ → parabole applatie

13) $h = 0\,m$ (simplification).

vitesse :

$\frac{dv_x}{dt} = a_x = 0 \Rightarrow V_x(t) = V_{0x} = V_0 \cos\alpha$

$\frac{dv_y}{dt} = a_y = -g \Rightarrow V_y(t) = \int -g\,dt$

$\Rightarrow V_y(t) = -gt + k$

$V_y(0) = k = V_{0y} = V_0 \sin\alpha$

$V_y(t) = V_0 \sin\alpha - gt$

$\vec{V} = V_0 \cos\alpha\, \vec{u_x} + (V_0 \sin\alpha - gt)\vec{u_y}$

position :

$\frac{dx}{dt} = V_0 \cos\alpha \rightarrow x(t) = V_0 \cos\alpha\, t + k$

$x(0) = k = 0 \Rightarrow x(t) = (V_0 \cos\alpha)t$

$\frac{dy}{dt} = V_0 \sin\alpha - gt \Rightarrow y(t) = \int (V_0 \sin\alpha - gt)\,dt$

$y(t) = V_0 \sin\alpha\, t - \frac{gt^2}{2} + h$

$y(0) = k = h = 0$

$y(t) = V_0 \sin\alpha\, t - \frac{gt^2}{2}$

$\vec{OM}(t) = (V_0 \cos\alpha\, t)\vec{u_x} + (V_0 \sin\alpha\, t - \frac{gt^2}{2})\vec{u_y}$

trajectoire (d'après 8))

$y = \frac{-g}{2V_0^2 \cos^2\alpha} x^2 + x \tan\alpha$

portée (d'après 10)) (**) = 0 car $h = 0$

$x_p = \frac{V_0^2 \cos\alpha \sin\alpha}{g} \left(1 + \sqrt{1 + 0}\right)$

$= \frac{2V_0^2 \cos\alpha \sin\alpha}{g} = \frac{V_0^2}{g} \sin 2\alpha$

(14) Maximiser la portée:

Le sinus est maximale si $2\alpha = \frac{\pi}{2}$

$\Rightarrow \alpha = \frac{\pi}{4}$.

Obs: $x'_p(\alpha) = \frac{2V_0^2}{g}\cos 2\alpha = 0$

$\Rightarrow 2\alpha = \frac{\pi}{2} \Rightarrow \alpha = \frac{\pi}{4}$

(15) hauteur maximale:

$y = ax^2 + bx + c$

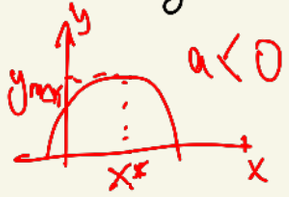
$y'(x^*) = 2ax^* + b = 0 \Rightarrow x^* = \frac{-b}{2a}$

pour le problème $a = \frac{-g}{2V_0^2\cos^2\alpha}$, $b = \tan\alpha$

Alternative 1:

$x^* = \frac{-\tan\alpha}{2\cdot\left(\frac{-g}{2V_0^2\cos^2\alpha}\right)} = \frac{V_0^2\cos^2\alpha\tan\alpha}{g}$

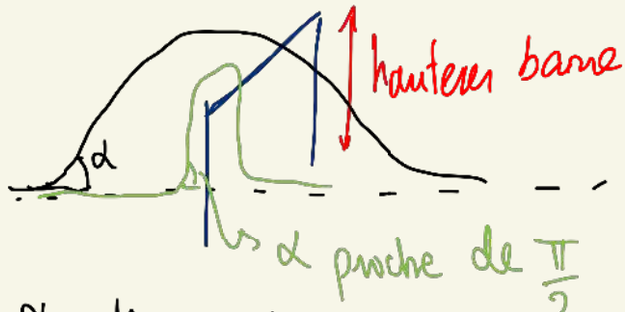
$= \frac{V_0^2\sin\alpha\cos\alpha}{g}$



(16) $y_{max} = \frac{V_0^2\sin^2\alpha}{2g}$

Le sinus est maximisé quand $\alpha = \frac{\pi}{2}$.

pour maximiser son hauteur l'athlète doit sauter le plus proche possible de la barre.



Obs: Un saut avec exactement $\alpha = \frac{\pi}{2}$ ne produit pas du mouvement horizontale. En pratique $\alpha \simeq \pi/2$.

$y_{max} = y(x^*) = \frac{-g}{2V_0^2\cos^2\alpha}\left(\frac{V_0^2\sin\alpha\cos\alpha}{g}\right)^2 + \tan\alpha\left(\frac{V_0^2\sin\alpha\cos\alpha}{g}\right)$

$= -\frac{V_0^2\sin^2\alpha}{2g} + \frac{V_0^2\sin^2\alpha}{g} = \frac{V_0^2\sin^2\alpha}{2g}$ //

Alternative 2:

$y_{max} = y(x^*) = a\left(\frac{-b}{2a}\right)^2 + b\left(\frac{-b}{2a}\right) + c$

$= \frac{b^2}{4a} - \frac{b^2}{2a} + c = -\frac{b^2}{4a} + c$

$y_{max} = \frac{-\tan^2\alpha}{4\cdot\left(\frac{-g}{2V_0^2\cos^2\alpha}\right)} = \frac{V_0^2(\tan\alpha\cos\alpha)^2}{2g} = \frac{V_0^2\sin^2\alpha}{2g}$